physics papers are immediately made available worldwide. The two papers described, from our widely different perspectives, the first examples of topology-changing transitions—the technical name for the space-tearing processes we had found. The long-standing question about whether the fabric of space can tear had been settled quantitatively by string theory.

Consequences

We have made much of the realization that spatial tears can occur without physical calamity But what does happen when the spatial fabric rips? What are the observable consequences? We have seen that many properties of the world around us depend upon the detailed structure of the curled-up dimensions. And so, you would think that the fairly drastic transformation from one Calabi-Yau to another as shown in Figure 11.5, would have a significant physical impact. In fact, though, the lower-dimensional drawings that we use to visualize the spaces make the transformation appear to be somewhat more complicated than it actually is. If we could visualize six-dimensional geometry, we would see that, yes, the fabric is tearing, but it does so in a fairly mild way. It’s more like the handiwork of a moth on wool than that of a deep knee bend on shrunken trousers.

Our work and that of Witten show that physical characteristics such as the number of families of string vibrations and the types of particles within each family are unaffected by these processes. As the Calabi-Yau space evolves through a tear, what can be affected are the precise values of the masses of the individual particles—the energies of the possible patterns of string vibrations. Our papers showed that these masses will vary continuously in response to the changing geometrical form of the Calabi-Yau component of space, some going up while others go down. Of primary importance, though, is the fact that there is no catastrophic jump, spike, or any unusual feature of these varying masses as the tear actually occurs. From the point of view of physics, the moment of tearing has no distinguishing characteristics.

This point raises two issues. First, we have focused on tears in the spatial fabric that occur in the extra six-dimensional Calabi-Yau component of the universe. Can such tears also occur in the more familiar three extended spatial dimensions? The answer, almost